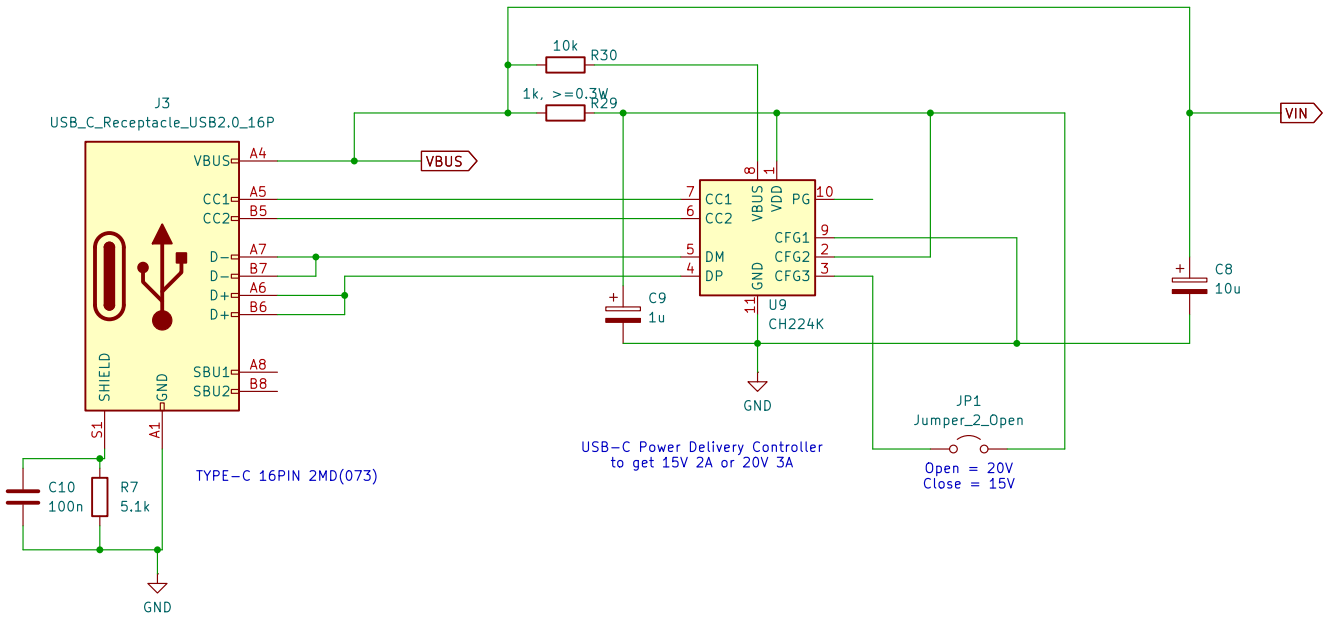
General USB-C based power supply:
- 5V 2A (minus what is drawn on 3.3V and 9V)
- 12V 1A
- -12V

Note:
- 12V is a linear regulator from Vin
- -5V is a linear regulator from -12V
- 3.3V and 9V are derived from 5V




1.4A:
- add 3.3V and 9V

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

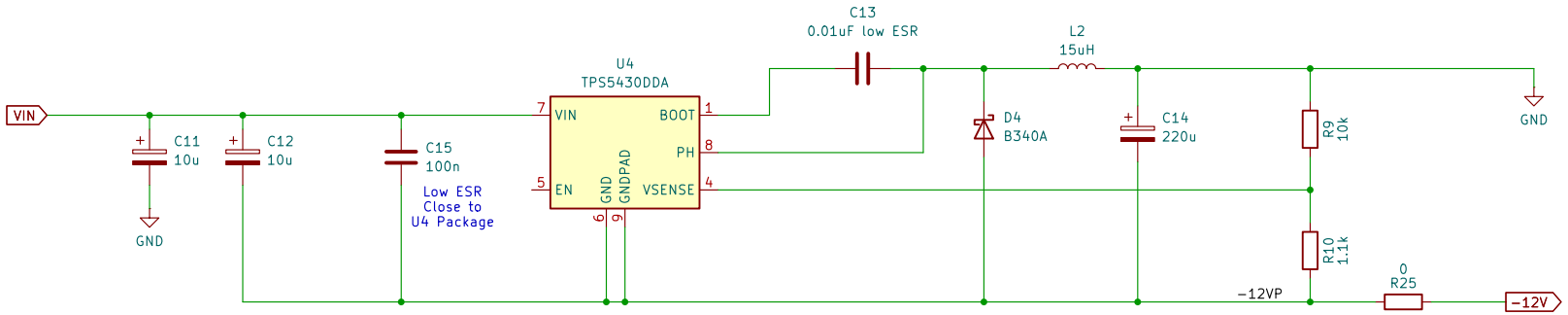
Power Detection and LEDs
File: pwrdetect.kicad_sch

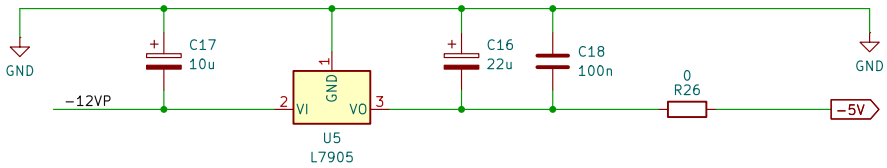
Positive Voltages
File: posvolt.kicad_sch

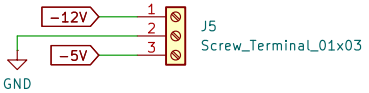
Negative Voltages
File: negvolt.kicad_sch

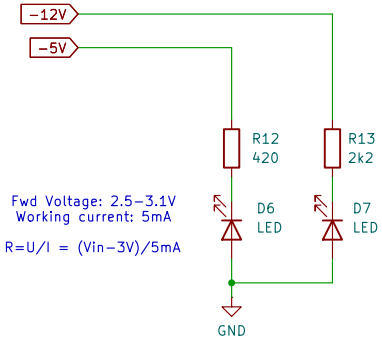
# USB-C Power Delivery Adapter 1.4A

USB-C-PWR-V1.4A-20260909



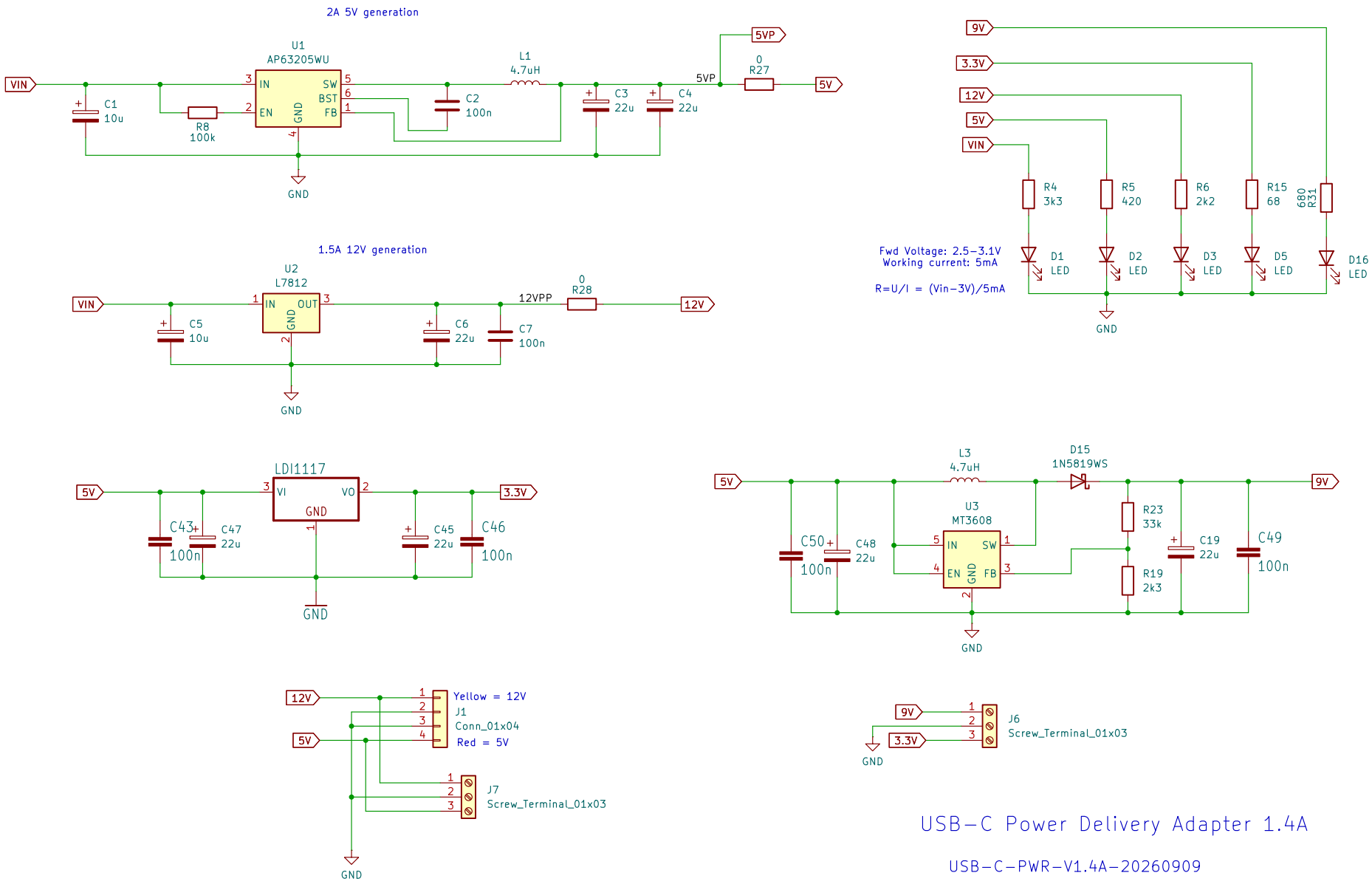




1.4A:
- proper terminal blocks

# USB-C Power Delivery Adapter 1.4A

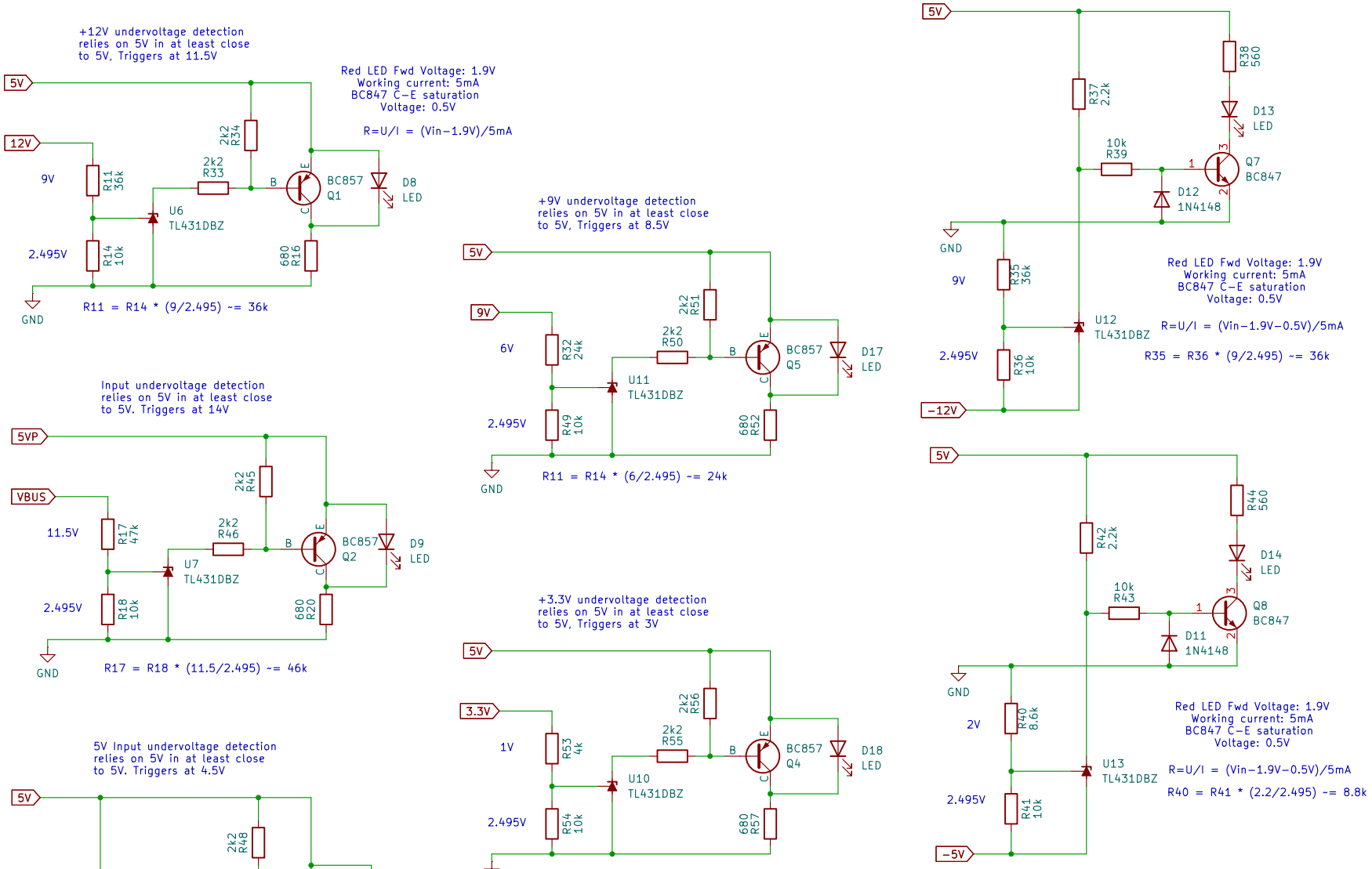
USB-C-PWR-V1.4A-20260909




# USB–C Power Delivery Adapter 1.4A

USB–C–PWR–V1.4A–20260909

1.4A:
– add 3.3V and 9V
– proper terminal blocks


+12V undervoltage detection
relies on 5V in at least close
to 5V, Triggers at 11.5V
Red LED Fwd Voltage: 1.9V
Working current: 5mA
BC847 C-E saturation
Voltage: 0.5V
R=U/I = (Vin-1.9V)/5mA
5V
12V
9V
R11
36k
2k2
R34
2k2
R33
B
BC857
Q1
E
C
D8
LED
U6
TL431DBZ
2.495V
R14
10k
680
R16
GND
R11 = R14 * (9/2.495) ~= 36k
+9V undervoltage detection
relies on 5V in at least close
to 5V, Triggers at 8.5V
5V
9V
6V
R32
24k
2k2
R51
2k2
R50
B
BC857
Q5
E
C
D17
LED
U11
TL431DBZ
2.495V
R49
10k
680
R52
GND
R11 = R14 * (6/2.495) ~= 24k
Input undervoltage detection
relies on 5V in at least close
to 5V. Triggers at 14V
5VP
VBUS
11.5V
R17
47k
2k2
R45
2k2
R46
B
BC857
Q2
E
C
D9
LED
U7
TL431DBZ
2.495V
R18
10k
680
R20
GND
R17 = R18 * (11.5/2.495) ~= 46k
+3.3V undervoltage detection
relies on 5V in at least close
to 5V, Triggers at 3V
5V
3.3V
1V
R53
4k
2k2
R56
2k2
R55
B
BC857
Q4
E
C
D18
LED
U10
TL431DBZ
2.495V
R54
10k
680
R57
GND
R11 = R14 * (6/2.495) ~= 4k
5V Input undervoltage detection
relies on 5V in at least close
to 5V. Triggers at 4.5V
5V
2V
R21
8.2k
2k2
R48
2k2
R47
B
BC857
Q3
E
C
D10
LED
U8
TL431DBZ
2.495V
R22
10k
680
R24
GND
R21 = R22 * (2/2.495) ~= 8k
5V
R37
2.2k
R38
560
D13
LED
10k
R39
1
3
2
Q7
BC847
D12
1N4148
GND
9V
R35
36k
Red LED Fwd Voltage: 1.9V
Working current: 5mA
BC847 C-E saturation
Voltage: 0.5V
U12
TL431DBZ
R=U/I = (Vin-1.9V-0.5V)/5mA
2.495V
R36
10k
R35 = R36 * (9/2.495) ~= 36k
-12V
5V
R42
2.2k
R44
560
D14
LED
10k
R43
1
3
2
Q8
BC847
D11
1N4148
GND
2V
R40
8.6k
Red LED Fwd Voltage: 1.9V
Working current: 5mA
BC847 C-E saturation
Voltage: 0.5V
U13
TL431DBZ
R=U/I = (Vin-1.9V-0.5V)/5mA
R40 = R41 * (2.2/2.495) ~= 8.8k
2.495V
R41
10k
-5V
USB-C Power Delivery Adapter 1.4A
USB-C-PWR-V1.4A-20260909
1.3B:
- fix brownout detection for
positive voltages
1.4A:
- add 3.3V and 9V